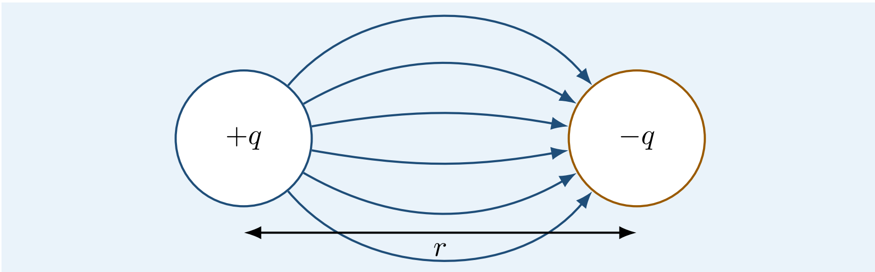
Electrostatics for Computer Science Students

A gentle mathematical introduction with examples

Lecture notes

May 17, 2026



Key idea

These notes are written for students who know programming better than physics. The goal is not to memorize formulas, but to learn how electrostatic formulas work, what the symbols mean, and how to use them without panic.

Contents

1 Why should a computer science student care?

Electrostatics is the part of electromagnetism that studies electric charges at rest. At first this may look far away from computer science. In practice it is everywhere in computing hardware.

- A transistor works because electric fields control the motion of charge carriers.
- A capacitor stores energy in an electric field. Memory cells, filters, touch screens and power supplies all use capacitance.
- Electrostatic discharge can destroy electronic components.
- Voltage is one of the most common physical quantities in electronics.
- Many ideas from electrostatics are mathematically similar to gravity, diffusion, optimization and potential functions.

Key idea

Think like a programmer: a charge creates a field, and the field is like a function that assigns a vector to every point in space.

$$\boldsymbol{E} : \mathbb{R}^3 \to \mathbb{R}^3.$$

At each point, the field tells a positive test charge which way it would be pushed.

2 Minimal mathematical toolbox

2.1 Scalars and vectors

A *scalar* has magnitude only. Examples: mass, charge, energy, temperature, electric potential.

A *vector* has magnitude and direction. Examples: force, velocity, electric field.

Quantity	Type	Unit
Charge q	scalar	coulomb, C
Force $\boldsymbol{F}$	vector	newton, N
Electric field $\boldsymbol{E}$	vector	N/C or V/m
Electric potential V	scalar	volt, V
Energy U	scalar	joule, J

2.2 Inverse-square laws

Many physical interactions in three-dimensional space have the form

$$\text{strength} \propto \frac{1}{r^2}.$$

If the distance doubles, the strength becomes four times smaller:

$$\frac{1}{(2r)^2} = \frac{1}{4r^2}.$$

If the distance triples, the strength becomes nine times smaller.

Common trap

Do not confuse $1/r$ with $1/r^2$. Electric potential from a point charge behaves like $1/r$. Electric field and electric force behave like $1/r^2$.

2.3 Superposition

Electrostatics is mostly linear. This means that if several charges produce fields, the total field is the vector sum of all fields:

$$\boldsymbol{E}_{\text{total}} = \boldsymbol{E}_1 + \boldsymbol{E}_2 + \cdots + \boldsymbol{E}_n.$$

This is called the *principle of superposition*.

In programming terms, if a single charge contributes one vector, then the total field can be computed by looping through all charges and adding their contributions.

Worked example

If three charges produce fields $\boldsymbol{E}_1 = (2, 0)\,\text{N/C}$, $\boldsymbol{E}_2 = (-1, 3)\,\text{N/C}$ and $\boldsymbol{E}_3 = (0, -4)\,\text{N/C}$ at the same point, then

$$\boldsymbol{E}_{\text{total}} = (2, 0) + (-1, 3) + (0, -4) = (1, -1)\,\text{N/C}.$$

3 Electric charge

Electric charge is a basic property of matter. It comes in two signs: positive and negative.

- Like charges repel: $+$ repels $+$, and $-$ repels $-$.
- Opposite charges attract: $+$ attracts $-$.
- The SI unit of charge is the coulomb, C.
- The elementary charge has magnitude

$$e \approx 1.602 \cdot 10^{-19}\,\text{C}.$$

An electron has charge $-e$, and a proton has charge $+e$.

3.1 Charge is quantized

A physical object cannot have an arbitrary charge. Its charge is a multiple of the elementary charge:

$$q = ne,$$

where n is an integer, possibly negative.

For macroscopic objects the number n is usually huge, so charge often looks continuous in practical calculations.

3.2 Charge is conserved

Electric charge is conserved. It can move from one object to another, but in an isolated system the total charge stays constant.

Key idea

When a body becomes positively charged, it usually did not create positive charge. It lost some electrons. When a body becomes negatively charged, it gained electrons.

4 Coulomb's law: force between two point charges

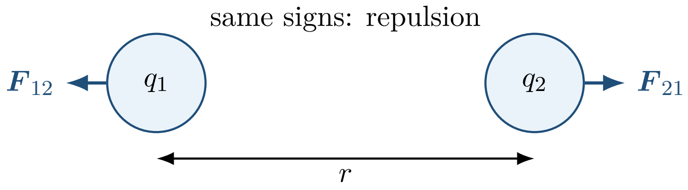
For two point charges q_1 and q_2 separated by distance r, the magnitude of the electrostatic force is

$$F = k_{\text{e}} \frac{|q_1 q_2|}{r^2},$$

where

$$k_{\text{e}} = \frac{1}{4\pi\varepsilon_0} \approx 8.99 \cdot 10^9\,\text{N}\,\text{m}^2/\text{C}^2.$$

Here ε_0 is the vacuum permittivity.



4.1 Vector form

The force is a vector. A useful vector form is

$$\boldsymbol{F}_{12} = k_\mathrm{e}\frac{q_1 q_2}{r^2}\,\hat{\boldsymbol{r}}_{12},$$

where $\hat{\boldsymbol{r}}_{12}$ is a unit vector pointing from charge q_1 to charge q_2, depending on the chosen convention. The sign of $q_1 q_2$ decides attraction or repulsion.

For beginners, it is often safer to do this in two steps:

1. Compute the magnitude using absolute values.
2. Decide direction from the signs of the charges.

Worked example

Two charges $q_1 = 2.0\,\mu\mathrm{C}$ and $q_2 = -3.0\,\mu\mathrm{C}$ are separated by $0.50\,\mathrm{m}$. Find the magnitude of the force.

$$F = k_\mathrm{e}\frac{|q_1 q_2|}{r^2} = 8.99 \cdot 10^9\,\frac{(2.0 \cdot 10^{-6})(3.0 \cdot 10^{-6})}{(0.50)^2}.$$

$$F = 8.99 \cdot 10^9\,\frac{6.0 \cdot 10^{-12}}{0.25} \approx 0.216\,\mathrm{N}.$$

The signs are opposite, so the force is attractive.

5 Electric field

The electric field is defined as force per unit positive test charge:

$$\boldsymbol{E} = \frac{\boldsymbol{F}}{q}.$$

Therefore

$$\boldsymbol{F} = q\boldsymbol{E}.$$

The unit of electric field is

$$\mathrm{N/C}.$$

It is also equal to V/m, which becomes clear when we introduce voltage.

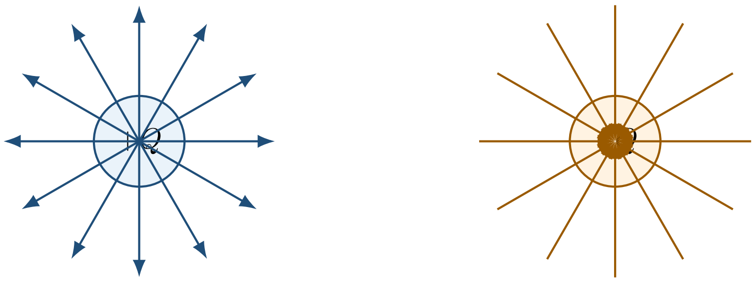
5.1 Field of a point charge

A point charge Q produces an electric field whose magnitude at distance r is

$$E = k_\mathrm{e}\frac{|Q|}{r^2}.$$

The direction is:

- away from a positive charge,
- toward a negative charge.



positive charge: field points outward negative charge: field points inward

5.2 Why the field concept is useful

Coulomb's law describes direct interaction between charges. The field approach separates the problem into two parts:

1. Source charges create an electric field.
2. A test charge placed in that field experiences a force.

This is like separating configuration from execution in software. First build the environment. Then ask what happens to an object placed inside it.

Worked example

At some point in space the electric field is $\boldsymbol{E} = (300, 0)\,\mathrm{N/C}$. What force acts on a charge $q = -2.0\,\mu\mathrm{C}$ placed there?

$$\boldsymbol{F} = q\boldsymbol{E} = (-2.0 \cdot 10^{-6})(300, 0) = (-6.0 \cdot 10^{-4}, 0)\,\mathrm{N}.$$

The negative charge is pushed opposite to the direction of the field.

6 Superposition of electric fields

If several charges are present, each creates its own field. The total field is the vector sum:

$$\boldsymbol{E} = \sum_{i=1}^{n} \boldsymbol{E}_i.$$

For a point charge q_i located at position $\boldsymbol{r}_i$, the field at point $\boldsymbol{r}$ is

$$\boldsymbol{E}_i(\boldsymbol{r}) = k_\mathrm{e} q_i \frac{\boldsymbol{r} - \boldsymbol{r}_i}{|\boldsymbol{r} - \boldsymbol{r}_i|^3}.$$

This formula may look scary, but it says something simple:

- $\boldsymbol{r} - \boldsymbol{r}_i$ points from the source charge to the observation point.
- The denominator gives the inverse-square distance behavior.
- The sign of q_i handles whether the field points away or toward the charge.

6.1 Algorithmic view

A simple numerical algorithm for the field at one point is:

```
E = (0, 0, 0)
for each charge qi at position ri:
    d = r - ri
    E += ke * qi * d / norm(d)^3
```

This is direct summation. It is not always the fastest method for huge systems, but it is conceptually correct.

Common trap

The electric field at the exact position of an ideal point charge is singular. In the formula, the distance is zero, so division by zero appears. This is a model limitation, not a programming bug.

7 Electric potential and voltage

Electric field is a vector. Electric potential is a scalar. Potential is often easier to add than fields because scalars do not have directions.

The electric potential V from a point charge Q at distance r is

$$V = k_\mathrm{e}\frac{Q}{r}.$$

The unit is the volt:

$$1\,\mathrm{V} = 1\,\mathrm{J/C}.$$

Key idea

Voltage is energy per unit charge. If a charge q moves through a potential difference ΔV, the change in electric potential energy is

$$\Delta U = q\Delta V.$$

7.1 Potential difference

Only potential differences matter physically. We usually write

$$\Delta V = V_B - V_A.$$

If $\Delta V = 5\,\mathrm{V}$, then each coulomb of positive charge gains $5\,\mathrm{J}$ of potential energy when moved from A to B.

7.2 Connection between field and potential

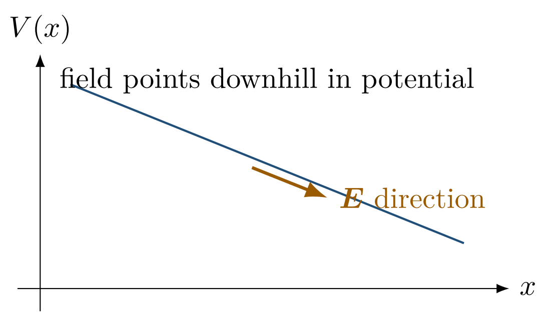
In one dimension,

$$E_x = -\frac{dV}{dx}.$$

In three dimensions,

$$\boldsymbol{E} = -\nabla V.$$

This means: the electric field points in the direction where potential decreases most rapidly.



8 Work and energy

The work done by an electric force when a charge moves from A to B is related to potential energy:

$$W_{\text{electric}} = -\Delta U.$$

Since $\Delta U = q\Delta V$, we have

$$W_{\text{electric}} = -q\Delta V.$$

For a uniform electric field and displacement d parallel to the field,

$$W = qEd.$$

Worked example

A proton moves through a potential difference of $\Delta V = -100\,\text{V}$. What is the change in its electric potential energy?

$$\Delta U = q\Delta V = (1.602 \cdot 10^{-19})(-100) = -1.602 \cdot 10^{-17}\,\text{J}.$$

The proton loses potential energy. The electric field can convert this energy into kinetic energy.

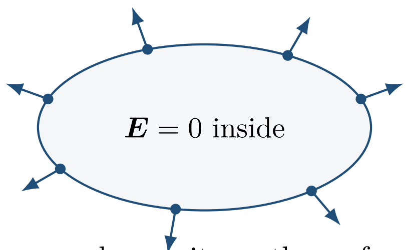
9 Conductors in electrostatic equilibrium

A conductor contains charges that can move. In electrostatic equilibrium, the charges have stopped rearranging. This gives several important facts.

1. The electric field inside a conductor is zero.
2. Excess charge lies on the surface.
3. The conductor is an equipotential object: every point on it has the same potential.
4. The electric field just outside the surface is perpendicular to the surface.

Key idea

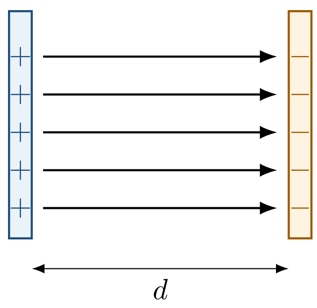
If there were a nonzero electric field inside a conductor, free charges would feel a force and move. Therefore equilibrium requires $\boldsymbol{E} = 0$ inside the conducting material.



excess charge sits on the surface

10 Capacitors

A capacitor is a device that stores separated charge and electric energy. The simplest model is two conducting plates separated by an insulator.



The capacitance is defined by

$$C = \frac{Q}{V},$$

where:

- Q is the magnitude of charge on one plate,
- V is the potential difference between the plates,
- C is capacitance, measured in farads, F.

For an ideal parallel-plate capacitor in vacuum,

$$C = \varepsilon_0 \frac{A}{d},$$

where A is plate area and d is plate separation.

The energy stored in a capacitor is

$$U = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{Q^2}{2C}.$$

Worked example

A capacitor has $C = 10\,\mu\text{F}$ and is charged to $V = 5\,\text{V}$. Find the stored energy.

$$U = \frac{1}{2}CV^2 = \frac{1}{2}(10 \cdot 10^{-6})(5^2) = 125 \cdot 10^{-6}\,\text{J}.$$

So $U = 1.25 \cdot 10^{-4}\,\text{J}$.

Common trap

A farad is a very large unit. In electronics you commonly see microfarads (μF), nanofarads (nF) and picofarads (pF).

11 Gauss's law

Gauss's law is a compact way to connect electric field and enclosed charge:

$$\oint_S \boldsymbol{E} \cdot d\boldsymbol{A} = \frac{Q_{\text{enc}}}{\varepsilon_0}.$$

The left side is electric flux through a closed surface. It measures how much electric field passes through the surface.

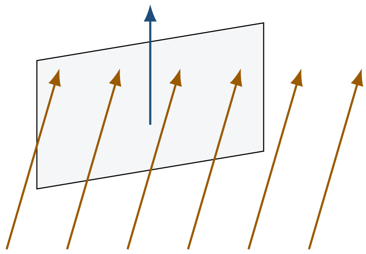
11.1 Flux intuition

If a field crosses a surface perpendicularly, it contributes strongly. If it is parallel to the surface, it contributes zero.

For a constant field through a flat surface,

$$\Phi_E = EA\cos\theta,$$

where θ is the angle between the field and the surface normal.



11.2 When Gauss's law is useful

Gauss's law is always true, but it is especially useful when the geometry has high symmetry:

- spherical symmetry,
- cylindrical symmetry,
- planar symmetry.

11.3 Example: field of a point charge

Choose a sphere of radius r around charge Q. By symmetry, the field has the same magnitude everywhere on the sphere and points radially outward. Therefore

$$\oint_S \boldsymbol{E} \cdot d\boldsymbol{A} = E \cdot 4\pi r^2.$$

Gauss's law gives

$$E \cdot 4\pi r^2 = \frac{Q}{\varepsilon_0}.$$

So

$$E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2} = k_{\mathrm{e}}\frac{Q}{r^2}.$$

This is the same result as Coulomb's law.

12 Uniform electric field

Between large parallel plates, the electric field is approximately uniform. This means it has the same magnitude and direction at many points.

If the potential difference between plates is V and the separation is d, then

$$E \approx \frac{V}{d}.$$

More precisely, the magnitude is $E = |\Delta V|/d$.

Worked example

Two plates are separated by $d = 2.0\,\mathrm{mm}$ and connected to a voltage of $12\,\mathrm{V}$. Estimate the field magnitude.

$$E = \frac{V}{d} = \frac{12}{2.0 \cdot 10^{-3}} = 6000\,\mathrm{V/m}.$$

13 Units and dimensional checks

Checking units catches many mistakes.

13.1 Coulomb's law units

$$F = k_e \frac{|q_1 q_2|}{r^2}.$$

The units are

$$\left(\mathrm{N\,m^2/C^2}\right)\frac{\mathrm{C^2}}{\mathrm{m^2}} = \mathrm{N}.$$

Correct: force should be in newtons.

13.2 Capacitor energy units

$$U = \frac{1}{2}CV^2.$$

Because $1\,\mathrm{F} = 1\,\mathrm{C/V}$,

$$\mathrm{F\,V^2} = \frac{\mathrm{C}}{\mathrm{V}}\mathrm{V^2} = \mathrm{C\,V} = \mathrm{J}.$$

Correct: energy should be in joules.

Key idea

Before trusting a numerical result, always ask: does the unit make sense? A force cannot have units of volts. An energy cannot have units of newtons.

14 Problem-solving recipes

14.1 Recipe 1: force between two charges

1. Convert charges to coulombs.
2. Convert distance to meters.
3. Use $F = k_e|q_1q_2|/r^2$.
4. Determine direction: same signs repel, opposite signs attract.

14.2 Recipe 2: force from electric field

1. Find or compute $\boldsymbol{E}$.
2. Use $\boldsymbol{F} = q\boldsymbol{E}$.
3. Remember: if $q < 0$, force is opposite to the field.

14.3 Recipe 3: potential energy

1. Identify the potential difference ΔV.
2. Use $\Delta U = q\Delta V$.
3. Interpret sign: positive means potential energy increases; negative means it decreases.

14.4 Recipe 4: capacitor calculations

1. Use $C = Q/V$ to connect charge, voltage and capacitance.
2. Use $U = \frac{1}{2}CV^2$ for stored energy.
3. Convert micro, nano and pico prefixes carefully.

15 Common mistakes

Common trap

Mistake 1: forgetting scientific notation. A microcoulomb is not 10^{-3} C but 10^{-6} C.

Common trap

Mistake 2: adding field magnitudes instead of vectors. If two fields point in opposite directions, they partially cancel.

Common trap

Mistake 3: thinking voltage is the same as electric field. Voltage is scalar energy per charge. Electric field is vector force per charge.

Common trap

Mistake 4: ignoring the sign of charge. A negative charge accelerates opposite to the electric field.

Common trap

Mistake 5: using Gauss's law mechanically without symmetry. Gauss's law is powerful, but solving for E from it is easy only when symmetry makes E constant on a chosen surface.

16 Worked examples

16.1 Example 1: electron in a uniform field

An electron is placed in a uniform electric field of magnitude $E = 2000\,\mathrm{N/C}$ pointing to the right. Find the magnitude and direction of the force on the electron.

The electron charge is

$$q = -e = -1.602 \cdot 10^{-19}\,\mathrm{C}.$$

The force is

$$\boldsymbol{F} = q\boldsymbol{E}.$$

The magnitude is

$$F = |q|E = (1.602 \cdot 10^{-19})(2000) = 3.204 \cdot 10^{-16}\,\mathrm{N}.$$

Because the charge is negative, the force points opposite to the field, so it points to the left.

16.2 Example 2: field halfway between equal positive charges

Two identical positive charges are placed at $x = -a$ and $x = a$. What is the electric field at $x = 0$?

Each charge produces a field of the same magnitude at the midpoint. The left positive charge pushes a positive test charge to the right. The right positive charge pushes it to the left. The two vectors cancel.

Therefore

$$\boldsymbol{E}(0) = \boldsymbol{0}.$$

This is not because there is no charge. It is because of symmetry and vector cancellation.

16.3 Example 3: potential from two charges

A charge $q_1 = 3\,\mu\text{C}$ is 0.20 m from a point P. A charge $q_2 = -1\,\mu\text{C}$ is 0.10 m from the same point. Find the electric potential at P.

Potential is scalar, so we add values directly:

$$V = k_e \frac{q_1}{r_1} + k_e \frac{q_2}{r_2}.$$

$$V = 8.99 \cdot 10^9 \left(\frac{3 \cdot 10^{-6}}{0.20} + \frac{-1 \cdot 10^{-6}}{0.10} \right).$$

$$V = 8.99 \cdot 10^9 \left(15 \cdot 10^{-6} - 10 \cdot 10^{-6} \right) = 8.99 \cdot 10^9 \cdot 5 \cdot 10^{-6}.$$

$$V \approx 4.50 \cdot 10^4\,\text{V}.$$

16.4 Example 4: capacitor charge

A 100 nF capacitor is connected to a 3.3 V source. Find the charge on one plate.

Use

$$Q = CV.$$

Convert

$$100\,\text{nF} = 100 \cdot 10^{-9}\,\text{F} = 1.00 \cdot 10^{-7}\,\text{F}.$$

Then

$$Q = (1.00 \cdot 10^{-7})(3.3) = 3.3 \cdot 10^{-7}\,\text{C}.$$

17 A tiny computational model

Electrostatics can be simulated numerically. For point charges, the electric field at a point is a sum of vector contributions.

```
import numpy as np

ke = 8.9875517923e9
charges = [
    (+1e-6, np.array([-1.0, 0.0, 0.0])),
    (-1e-6, np.array([+1.0, 0.0, 0.0])),
]

r = np.array([0.0, 1.0, 0.0])
E = np.zeros(3)

for q, ri in charges:
    d = r - ri
    E += ke * q * d / np.linalg.norm(d)**3

print(E)
```

Key idea

This code is just the formula

$$\boldsymbol{E}(\boldsymbol{r}) = \sum_i k_e q_i \frac{\boldsymbol{r} - \boldsymbol{r}_i}{|\boldsymbol{r} - \boldsymbol{r}_i|^3}.$$

A formula and a program are often two representations of the same algorithm.

18 Formula sheet

Meaning	Formula
Coulomb force magnitude	$F = k_\mathrm{e}\dfrac{\lvert q_1 q_2\rvert}{r^2}$
Electric field definition	$\boldsymbol{E} = \dfrac{\boldsymbol{F}}{q}$
Force in electric field	$\boldsymbol{F} = q\boldsymbol{E}$
Field of point charge	$E = k_\mathrm{e}\dfrac{\lvert Q\rvert}{r^2}$
Potential of point charge	$V = k_\mathrm{e}\dfrac{Q}{r}$
Potential energy change	$\Delta U = q\Delta V$
Field-potential relation	$\boldsymbol{E} = -\nabla V$
Capacitance	$C = \dfrac{Q}{V}$
Parallel-plate capacitor	$C = \varepsilon_0\dfrac{A}{d}$
Capacitor energy	$U = \dfrac{1}{2}CV^2 = \dfrac{1}{2}QV = \dfrac{Q^2}{2C}$
Gauss's law	$\oint_S \boldsymbol{E}\cdot d\boldsymbol{A} = \dfrac{Q_\mathrm{enc}}{\varepsilon_0}$
Uniform field between plates	$E \approx \dfrac{V}{d}$

19 Practice problems

1. Two charges $q_1 = 4\,\mu\mathrm{C}$ and $q_2 = 2\,\mu\mathrm{C}$ are separated by $0.30\,\mathrm{m}$. Find the magnitude of the force and decide whether it is attractive or repulsive.
2. A charge $q = -5\,\mu\mathrm{C}$ is placed in an electric field $\boldsymbol{E} = (100, 0)\,\mathrm{N/C}$. Find the force.
3. A point charge $Q = 1.0\,\mu\mathrm{C}$ creates an electric field. Find the field magnitude at $r = 0.50\,\mathrm{m}$.
4. A point charge $Q = -2.0\,\mu\mathrm{C}$ is located $0.25\,\mathrm{m}$ from point P. Find the electric potential at P.
5. A capacitor has capacitance $C = 47\,\mu\mathrm{F}$ and voltage $V = 12\,\mathrm{V}$. Find the charge on one plate.
6. A $220\,\mathrm{nF}$ capacitor is charged to $9\,\mathrm{V}$. Find the stored energy.
7. The voltage between two plates is $24\,\mathrm{V}$ and their separation is $3.0\,\mathrm{mm}$. Estimate the electric field magnitude.
8. Explain in one sentence why the electric field inside a conductor is zero in electrostatic equilibrium.
9. Two equal positive charges are placed symmetrically around the origin. What can you say about the electric field at the origin?
10. A charge moves through a potential difference of $\Delta V = 20\,\mathrm{V}$. If $q = 3.0\,\mu\mathrm{C}$, find ΔU.

20 Answers

1. $$F = 8.99\cdot 10^9\,\frac{(4\cdot 10^{-6})(2\cdot 10^{-6})}{0.30^2} \approx 0.80\,\mathrm{N}.$$
   Same signs, so the force is repulsive.
2. $$\boldsymbol{F} = q\boldsymbol{E} = (-5\cdot 10^{-6})(100, 0) = (-5\cdot 10^{-4}, 0)\,\mathrm{N}.$$
3. $$E = 8.99\cdot 10^9\,\frac{1.0\cdot 10^{-6}}{0.50^2} \approx 3.60\cdot 10^4\,\mathrm{N/C}.$$

4. $$V = 8.99 \cdot 10^9 \frac{-2.0 \cdot 10^{-6}}{0.25} \approx -7.19 \cdot 10^4\,\mathrm{V}.$$
5. $$Q = CV = (47 \cdot 10^{-6})(12) = 5.64 \cdot 10^{-4}\,\mathrm{C}.$$
6. $$U = \frac{1}{2}CV^2 = \frac{1}{2}(220 \cdot 10^{-9})(9^2) \approx 8.91 \cdot 10^{-6}\,\mathrm{J}.$$
7. $$E = \frac{V}{d} = \frac{24}{3.0 \cdot 10^{-3}} = 8.0 \cdot 10^3\,\mathrm{V/m}.$$
8. If the field inside a conductor were nonzero, free charges would move, so the conductor would not be in electrostatic equilibrium.
9. The field at the origin is zero because the two field vectors have equal magnitude and opposite directions.
10. $$\Delta U = q\Delta V = (3.0 \cdot 10^{-6})(20) = 6.0 \cdot 10^{-5}\,\mathrm{J}.$$

21 Final summary

- Charge creates electric field.
- Electric field creates force on charge: $\boldsymbol{F} = q\boldsymbol{E}$.
- Electric potential is energy per charge: $V = U/q$.
- Potential differences matter more than absolute potential values.
- Capacitors store charge and energy in electric fields.
- Gauss's law is a global relation between flux and enclosed charge.
- Units and signs are not decoration; they are part of the calculation.

Key idea

The most important mental model is simple: electrostatics is about invisible vector fields created by charge. Once you understand the field, forces, energy and voltage become much less mysterious.